Axel Refalo

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Passionate about robotics, I study the morphology of serial manipulators, what the arrangement of the joints implies for industrial performance. As robots grow increasingly complex, controllability becomes a question that has to be answered at the design stage. I am currently pursuing my master's thesis under the supervision of Prof. Ilian Bonev and Prof. Clément Gosselin.

Through this research I have analysed ABB's GoFa and FANUC's CRX series, and released an inverse-kinematics solver that makes such analysis possible. I also build the robotics hardware: a plant-imaging gantry at McGill, and two Cartesian inspection robots at Canon Production Printing. The designs, source code, papers and documentation behind each of these projects are available at `axelrefalo.github.io`.

Research interests: mechanisms in robotics, experimental robotics and machine learning

EDUCATION

École de Technologie Supérieure (ÉTS) — Montréal, Canada
M.Sc.A. in Automation Engineering, research-based | GPA 3.95/4.30 — *Sept. 2024 – Sept. 2026*

- Thesis: *Singularity Analysis and Path Planning Algorithm Design*, Control and Robotics Laboratory. Supervisor: Prof. Ilian Bonev; co-supervisor: Prof. Clément Gosselin
- Coursework in advanced control

McGill University — Montréal, Canada
Graduate coursework in Computer Science, exchange program | GPA 4.00/4.00 — *Sept. 2024 – May 2025*

- *Applied Robotics* – ROS controller packages for the Kinova Gen3 seven-axis arm, validated in Gazebo, implemented in C++. *Ranked first*
- *Applied Machine Learning* – pattern recognition and classification: regression, decision trees, neural networks and transformers in Python

University of Technology of Belfort-Montbéliard (UTBM) — Belfort, France
B.Eng. and M.Eng. in Mechanical Engineering, specialisation in Mechatronics — *Sept. 2020 – Feb. 2026*

- Five-year integrated engineering programme, completed as a double degree with ÉTS
- Coursework: Control systems, continuum mechanics, finite element methods, CAD and additive manufacturing; electronic design and signal processing; scientific computing in C++, C# and LabVIEW

University of Palermo — Palermo, Italy
Mechanical Engineering, exchange program — *Sept. 2021 – Feb. 2022*

- Kinematics and dynamics of mechanical systems, taught in Italian

PAPERS

Peer-reviewed conference papers

[ICRA 2026 Conference] **Refalo, A.**, Bonev, I., and Gosselin, C. "Singularity Analysis of ABB's GoFa 5 Robot Arm." *Proceedings of the IEEE International Conference on Robotics and Automation*, Vienna, Austria, June 2026.

Derives the complete singularity loci in the joint space using Grassmann line geometry, and compares them to those of conventional morphologies.

Manuscripts in preparation

[ICRA 2027 Conference] **Refalo, A.**, Mac Leod, P., Bonev, I., and Gosselin, C. "Singularity Analysis of FANUC's CRX Series." *In preparation for submission to the IEEE International Conference on Robotics and Automation, Seoul, South Korea, May 2027.*

Derives the complete singularity loci using the methodology of [ICRA 2026], and shows the discrepancies between the controller model and the actual kinematics of the arm.

[Robotics Journal] **Refalo, A.**, Bonev, I., and Gosselin, C. "Morphology Comparison of Robot Arms for Industrial Tasks." *In preparation for submission to Robotics MDPI journal, September 2026.*

Study whether unconventional morphologies outperform conventional ones, and whether the added complexity is justified.

[Plant Methods Journal] Debbagh, M., **Refalo, A.**, *et al.* "An Affordable and Open-Source Gantry Framework for Capturing High-Resolution Spatiotemporal Plant Growth Data." *In preparation for submission to Plant Methods journal, October 2026.*

Designed and built the gantry platform: mechanical design, motion control and sensor acquisition.

RESEARCH EXPERIENCE

I have collaborated with laboratories across the span of my interests, sometimes alone on a project, sometimes as part of a team. Most of this research, along with conference and travel costs, was funded by the Fonds de recherche du Québec.

Control and Robotics Laboratory (CoRo) ÉTS, Montréal
Graduate researcher, M.Sc. thesis – Prof. Ilian Bonev, with Prof. Clément Gosselin *May 2025 – present*

The arrangement of a robot's joints determines its inverse kinematics and its singular loci. I explore how morphology governs performance, by identifying the singular loci and the practical behaviour of the arm around them.

- Released IKDH, an inverse-kinematics solver returning *all* inverse kinematics solutions for a general six-revolute arm from its Denavit–Hartenberg parameters, where industrial software returns at most eight. Written in Python; used at CoRo and ÉTS
- Singularity analysis deriving and comparing the singular loci of collaborative arms from ABB, FANUC, Universal Robots and Doosan. Validation in RobotStudio and Roboguide, revealing discrepancies between the controller models and the true kinematics [ICRA 2026, ICRA 2027]
- Attended the 5$^{\text{th}}$ Summer School on Singularities of Mechanisms and Robotic Manipulators, Barcelona, 2025

Bioresource Engineering Laboratory McGill, Montréal
Research assistant – PhD Mohamed Debbagh *Jan. 2025 – present*

Modelling how plants grow in space and time requires dense, repeatable imaging. Commercial phenotyping gantries are expensive and closed-source, which puts them out of reach for laboratories.

- Designed, built and programmed a five-axis imaging gantry from scratch – three linear, two rotational – covering mechanics, drive electronics, motion control and sensor acquisition
- Interdisciplinary collaboration with plant scientists and roboticists, who use the resulting time series to model spatiotemporal growth patterns

Materials Science Laboratory ÉTS, Montréal
Research project – Prof. Claudiane Ouellet-Plamondon *Dec. 2024 – May 2025*

Synchrotron tomography can resolve the microstructure of cementitious materials, but raw projections are too noisy to segment reliably.

- Built a computed-tomography reconstruction and filtering pipeline for synchrotron scans of concrete, turning raw projections into volumes clean enough for quantitative porosity and crack analysis with Python and Dragonfly

Laboratory of Space Technologies ÉTS, Montréal
Research project – Prof. René Landry *Sept. 2024 – Dec. 2024*

Geolocation using the Doppler shift of signals of opportunity from low Earth orbit satellites, locating the receiver without communicating directly with the satellite.

- Implemented a Kalman filter estimating receiver position from signals of opportunity emitted by low Earth orbit satellite constellations

Canon Production Printing R&D Eindhoven, Netherlands
Internship – Dr. Karel Knechten *Sept. 2023 – Feb. 2024*

Large-format print quality is critical, and it is threatened by ink contamination and nozzle failure. Automated tools that capture this fast, microscopic behaviour give the research team the insights it needs to develop more reliable printers.

- Developed two Cartesian inspection robots for large-format printers: (I) imaging and spectrometric analysis of microscopic nozzle defects and contamination; (II) high-speed microscopic imaging of ink behaviour on the coating surface
- Responsible for the mechanical and electronic design, the motion control for spectrometric measurement and microscopic imaging, and the user interface through which the team operates them

LEADERSHIP

FabLab Manager — Design Department, ÉTS — *Aug. 2025 – Dec. 2025*

- Supervised students on additive manufacturing, laser cutting and electronics. Upgraded the workshop and created a wiki for documentation and workflow

Research intern supervision — CoRo, ÉTS — *May 2025 – Sept. 2025*

- Supervised an intern over five months on the inverse kinematics of the FANUC CRX series

CrunchTime hackathon – participant, then electronics manager — UTBM — *2022 – 2024*

Crunch industry camp – finalist

ENGINEERING PROJECTS

Independent projects in which I designed the mechanism, the electronics and the control code myself.

Robotics

- Two-cable-driven parallel plotter for spray painting — 2026
- Three-degree-of-freedom manipulator with parallelogram mechanism — 2024
- Wide planar plotter driven differentially — 2022

Computer Aided Design

- Mechanical and electronic design of all the robotics projects above, built from scratch
- Replication of a Swiss watch mechanism – parametric gears with a cycloidal profile — 2023

Programming

- All the robotics projects above, coded mostly in C++ or Python — 2025
- Simple Eulerian fluid simulator in C++ — 2023
- Algorithms to solve complex grid puzzles using backtrack search in C++ — 2023
- Mechanical truss simulator using finite element methods in C# — 2022

SKILLS

Languages French (native), English (C1 – IELTS 7.0), Italian (fluent), Spanish (beginner)

Software *Robotics*: Robot Operating System, Gazebo, RoboDK, RobotStudio and Roboguide
Programming: Python, C++, C#, LabVIEW, Unix shell, Git, LaTeX and Visual Studio Code
Math: MATLAB, SciPy, Wolfram, Maple and GeoGebra

Mechanics *Computer Aided Design*: CATIA V5, Fusion 360 and AutoCAD
Physical modeling: Simulink, Amesim, Ansys, OrCAD, Abaqus and Dragonfly

Electronics Controller boards (Raspberry Pis and Arduinos), motors (stepper and brushless), batteries and power converters, and usual electronic components

Manufacturing Additive manufacturing, laser cutter, electronic soldering and hand tools

REFERENCES

Prof. Ilian Bonev
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Prof. Claudiane Ouellet-Plamondon
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